Journal of Computational and Applied Mathematics 440 (2024) 115662

Contents lists available at ScienceDirect

Journal of Computational and Applied Mathematics

journal homepage: www.elsevier.com/locate/cam

Analytically pricing European options with a two-factor Stein–Stein model

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ARTICLE INFO

Keywords:
Stein–Stein model
European options
Stochastic mean reversion
Analytical solution
Empirical study

ABSTRACT

Stochastic volatility models are widely used in option pricing; however, most of these models assume a constant long-term mean of volatility that fails to account for the implied volatility term structure. In response, this article proposes an improved Stein–Stein model that enables the random variation of the long-term mean under the Stein–Stein model while also relaxing the zero correlation assumption between asset and volatility. One of the new model's chief advantages is that it offers an analytical solution to European option prices, thus speeding up the calibration process and allowing for easy implementation in practical risk management. An empirical analysis based on 50ETF options in the Chinese market demonstrates that the newly improved Stein–Stein model predicts market outcomes more accurately compared to the original Stein–Stein model.

1. Introduction

Options play a crucial role in risk diversification for investors as a financial derivative. Scholars, such as Bachelier [1], Boness [2], and Samuelson [3], studied a range of option pricing models before 1973. In 1973, Black and Scholes [4] noted that these models failed to consider investors' demands for asset diversity, resulting in a discount rate that factored in the risk premium investors demanded for holding both stock prices and options. Furthermore, these models relied on the typical investor's utility function, making the models difficult to apply universally. Even today, the model proposed by Black and Scholes (BS model) is a widely employed option pricing model because of its simplicity and pricing efficiency. Nevertheless, the BS model has limitations due to its strict assumptions, making it biased towards the pricing of European-style options [5] and unable to account for realistic phenomena such as the "volatility smile", "leverage effect", and underlying asset discontinuity [6].

Stochastic volatility (SV) models have been developed to better explain the phenomenon of the "volatility smile" and account for time-varying volatility. One notable example is the class of single-factor stochastic volatility models. By utilizing a numerical solution to price options, Wiggins [7] demonstrated that the SV model outperforms the Black–Scholes (BS) model in valuing in-the-money call options with longer maturities. Hull and White [5] made a further improvement by deriving a semi-analytical solution when the underlying asset followed another SV model. Through market data analysis, they confirmed that their model provided more accurate pricing than the BS model, with the difference between the two models increasing as the option expiration lengthens. Stein and Stein [8] proposed an Ornstein–Uhlenbeck (OU) process for volatility in their stochastic volatility (SV) model and obtained an analytical solution for the Stein–Stein model when the underlying asset is not correlated with volatility. Although this model has a smaller pricing bias for options than the Black–Scholes (BS) model, it cannot capture the "leverage effect" phenomenon. Schöbel

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https://doi.org/10.1016/j.cam.2023.115662
Received 15 April 2023; Received in revised form 10 September 2023
Available online 3 November 2023
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and Zhu [9] incorporated a non-zero correlation between the underlying asset and volatility to derive an improved model with an analytical solution. Hozman and Tich [10] proposed a modified Stein–Stein model that ensures non-negativity of volatility by taking the absolute value of the diffusion term of the log price. They applied the discontinuous Galerkin method to the initial boundary conditions of European option prices to obtain numerical solutions. He and Zhu [11] developed an alternative form of the Stein–Stein model to reduce pricing biases in options. Additionally, Heston [12] modeled asset volatility using the Cox–Ingersoll–Ross (CIR) process, which is correlated to the asset, and the model admits an analytic solution.

Bates [10] introduced jumps in the underlying asset price of a single-factor volatility model, while Eraker [13] extended this approach by allowing for jumps in volatility as well. Empirically, however, Eraker found that including the jump term did not significantly improve the biases of in-sample option prices. Sepp [14] derived an analytical solution for swap options by incorporating jumps in asset values and their volatilities. Kallsen et al. [15] studied the case of stock prices following a time-varying Lévy process and derived a semi-explicit price formulation under the one-factor stochastic volatility model. Most recently, Jiao et al. [16] added a self-excited jump structure to the asset variance of a single-factor stochastic volatility model, enabling the model to explain the phenomenon of jump aggregation.

While single-factor SV models have some limitations in capturing the term structure of volatility, Duffie et al. [17] proposed an affine formulation treatment for a class of two-factor SV models to overcome this issue. As a result, multi-factor SV models have become increasingly popular. Gallant et al. [18] empirically demonstrated that a three-factor model fits better than a two-factor one for S&P 500 options. Bardgett et al. [19] improved the pricing ability of a two-factor model by adding a jump term, while Chang and Wang [20] replaced the Brownian part in the two-factor Heston model with a fractional Brownian motion to enhance pricing performance. To reflect macroeconomic influences, Lin and He [21] introduced a regime-switching-factor to volatility fluctuations in the SV model and verified empirically the smaller pricing bias of the new model after deriving an analytical solution for European options. Elliott and Lian [22] replaced the long-run mean of volatility with a regime-switching factor, while the concept of a stochastic long-term mean for stochastic volatility was introduced by He and Chen [23] to form a two-factor SV model. The combination of these two models has resulted in a three-factor SV model [24].

A new two-factor stochastic volatility model is presented in this paper that improves upon the Stein–Stein model and provides a better explanation for the implied volatility term structure. In addition to allowing for correlated asset and volatility, we relax the assumption that the long-term mean of volatility is constant under the Stein–Stein model. We show tat European options can be valued via a derived closed-form solution, followed by numerical experiments and an empirical study.

The main contribution of this paper can be summarized from two aspects. Firstly, the proposed model admits a closed-form solution to European option prices, which guarantees the efficiency in practice. More importantly, the proposed model is shown to be able to capture the implied volatility term structure much better than the classical Stein–Stein model does, which means that it is able to provide valuable insights for option pricing and risk management. The new model is also shown to be able to given an overall improvement of 16% in both in- and out-of-sample errors, compared with the classical Stein–Stein model, and such improvement is significant at a 1% significance level.

Here is the organization of the remaining parts. Section 2 introduces the construction of the new model from the Stein–Stein model. Section 3 numerically analyzed the new model on the correctness and the impact of relevant parameters on option pricing. Section 4 uses market data to show that the revised model outperforms the Stein–Stein model for option pricing, with the conclusion provided in the last section.

2. European option pricing

2.1. The Stein–Stein model

The Stein–Stein model assumes that the underlying price obeys the following stochastic differential equation:

$$\begin{aligned} \frac{dS_t}{S_t} &= rdt + v_t dW_t^1, \\ dv_t &= k(\overline{v} - v_t)dt + \sigma dW_t^2 \end{aligned} \tag{1}$$

where $\{S_t, t \geq 0\}$ and $\{v_t, t \geq 0\}$ are used to represent the price and volatility of the underlying asset, respectively. v_t is variance, whose volatility and long-term average value are denoted by σ and $\overline{v}$, respectively. k denotes the rate returning to this long-term average value. r refers to the risk-free rate. Additionally, W_t^1 and W_t^2 are both Brownian motions, and are uncorrelated with each other.

2.2. The new model

This paper aims to enhance the Stein–Stein model's capability to explain the term structure of implied volatility and variance swap curves by introducing the θ_t parameter. θ_t denotes the long-run mean value of volatility at a specific time interval t. In addition, we relaxed the assumption of the Stein–Stein model that volatility is independent of the underlying asset and developed an improved SV model, which we will refer to as the "new model".

The new model's differential equation governing the price of the underlying asset is given by:

$$\frac{dS_t}{S_t} = rdt + v_t dW_t^1,$$

$$\begin{aligned} dv_t &= k(\theta_t - v_t)dt + \sigma dW_t^2, \\ d\theta_t &= \lambda dt + \sigma_2 dB_t. \end{aligned} \tag{2}$$

It is assumed that the added Geometric Brownian motion B_t is uncorrelated with both W_t^1 and W_t^2. However, the assumption that $\rho = 0$ cannot explain the leverage effect of volatility and return [25]. Therefore, the new model allows for W_t^1 and W_t^2 to be correlated with two correlation coefficients with $|\rho| \le 1$.

The new expression for the long-term mean of volatility indicates that when λ and σ_2 are small, $d\theta_t$ approaches zero, resulting in the long-term mean of volatility being approximated as a constant value. If $\rho = 0$, the new model converges to the Stein–Stein model. However, when the value of ρ deviates from 0, or the long-term mean of volatility varies over time (λ or σ_2 deviates from 0), the new model differs from the Stein–Stein model. Although the introduction of a stochastic long-term mean adds another dimension to the target problem, we are still able to derive an analytical solution, which is presented in the following theorem.

Theorem 2.1. *If the underlying dynamics are given in Eq. (2), one can derive an analytical form of European option prices as*

$$U(S_t, v_t, \theta_t, t) = S_t P_1 - K e^{-r(T-t)} P_2 \tag{3}$$

where

$$\begin{aligned} P_j &= \frac{1}{2} + \frac{1}{\pi}\int_0^\infty RE\left[\frac{e^{-i\phi\ln(K)}}{i\phi} f_j\right] d\phi, \\ f_j &= e^{C(\tau;\phi)+D(\tau;\phi)v_t^2+E(\tau;\phi)v\theta_t+F(\tau;\phi)\theta_t^2+G(\tau;\phi)\theta_t+H(\tau;\phi)v+i\phi x}, \end{aligned}$$

The structure of the characteristic functions in affine-diffusion models is an exponential in the state variables. see Lord and Kahl's paper

with $C(\tau;\phi)$, $D(\tau;\phi)$, $E(\tau;\phi)$, $F(\tau;\phi)$, $G(\tau;\phi)$ and $H(\tau;\phi)$ satisfying

$$\begin{aligned} \frac{\partial D(\tau;\phi)}{\partial \tau} &= (u_j i\phi - \frac{1}{2}\phi^2)\tau + 2\sigma^2 D^2(\tau;\phi) - 2A(\phi)D(\tau;\phi) + \frac{1}{2}\sigma_2^2 E^2(\tau;\phi), \\ \frac{\partial E(\tau;\phi)}{\partial \tau} &= 2kD(\tau;\phi) - A(\phi)E(\tau;\phi) + 2\sigma^2 D(\tau;\phi)E(\tau;\phi) + 2\sigma_2^2 E(\tau;\phi)F(\tau;\phi), \\ \frac{\partial F(\tau;\phi)}{\partial \tau} &= kE(\tau;\phi) + \frac{1}{2}\sigma^2 E^2(\tau;\phi) + 2\sigma_2^2 F^2(\tau;\phi), \\ \frac{\partial G(\tau;\phi)}{\partial \tau} &= kH(\tau;\phi) + 2\lambda F(\tau;\phi) + \sigma^2 E(\tau;\phi)H(\tau;\phi) + 2\sigma_2^2 F(\tau;\phi)G(\tau;\phi), \\ \frac{\partial H(\tau;\phi)}{\partial \tau} &= -A(\phi)H(\tau;\phi) + \lambda E(\tau;\phi) + 2\sigma^2 D(\tau;\phi)H(\tau;\phi) + \sigma_2^2 E(\tau;\phi)G(\tau;\phi), \\ \frac{\partial C(\tau;\phi)}{\partial \tau} &= ri\phi + \lambda G(\tau;\phi) + \frac{1}{2}\sigma^2 H^2(\tau;\phi) + \sigma^2 D(\tau;\phi) + \frac{1}{2}\sigma_2^2 G^2(\tau;\phi) + \sigma_2^2 F(\tau;\phi), \end{aligned} \tag{4}$$

Here, $j = 1, 2, u_1 = \frac{1}{2}, b_1 = 1, u_2 = -\frac{1}{2}, b_2 = 0, A(\phi) = k - \sigma\rho(b_j + i\phi)$.

Proof. Since $U(S_t, v_t, \theta_t, t)$ is a function of S_t, v_t and t, the generalized Ito's Lemma implies

$$\begin{aligned} dU(S_t, v_t, \theta_t, t) &= (rS_t dt + v_t S_t dW_t^1)\frac{\partial U(S_t, v_t, \theta_t, t)}{\partial S} + (k(\theta_t - v_t)dt + \sigma dW_t^2)\frac{\partial U(S_t, v_t, \theta_t, t)}{\partial v_t} \\ &\quad + (\lambda dt + \sigma_2 dB_t)\frac{\partial U(S_t, v_t, \theta_t, t)}{\partial \theta_t} + \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial t} + \frac{1}{2}v_t^2 S_t^2 \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial S_t^2} dt \\ &\quad + \frac{1}{2}\sigma^2 \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial v_t^2} dt + \frac{1}{2}\sigma_2^2 \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial \theta_t^2} dt + \sigma v_t S_t \rho \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial S_t \partial v_t} dt \\ &= (\frac{1}{2}v_t^2 S_t^2 \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial S_t^2} + \frac{1}{2}\sigma^2 \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial v_t^2} + \frac{1}{2}\sigma_2^2 \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial \theta_t^2} \\ &\quad + \sigma v_t S_t \rho \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial S_t \partial v_t} + rS_t \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial S_t} + \lambda \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial \theta_t} \\ &\quad + k(\theta_t - v_t)\frac{\partial U(S_t, v_t, \theta_t, t)}{\partial v_t} + \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial t})dt + \sigma \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial v_t} dW_t^2 \\ &\quad + v_t S_t \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial S} dW_t^1 + \sigma_2 \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial \theta_t} dB_t \end{aligned} \tag{5}$$

We can then obtain

$$\begin{aligned} d(e^{-rt}U(S_t, v_t, \theta_t, t)) &= e^{-rt}(-rU(S_t, v_t, \theta_t, t) + \frac{1}{2}v_t^2 S_t^2 \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial S_t^2} + \frac{1}{2}\sigma^2 \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial v_t^2} \\ &\quad + \frac{1}{2}\sigma_2^2 \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial \theta_t^2} + \sigma v_t S_t \rho \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial S_t \partial v_t} + rS_t \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial S_t} \\ &\quad + k(\theta_t - v_t)\frac{\partial U(S_t, v_t, \theta_t, t)}{\partial v_t} + \lambda \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial \theta_t} \end{aligned} \tag{6}$$

$$+ \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial t})dt + e^{-rt}(v_t S_t \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial S_t} dW_t^1$$
$$+ \sigma \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial v_t} dW_t^2 + \sigma_2 \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial \theta_t} dB_t).$$

Since the option price is a martingale under the premise of risk neutrality, the price $U(S_t, v_t, \theta_t, t)$ of the European call option should satisfy the following partial differential equation

$$\begin{aligned}
&\frac{1}{2} v_t^2 S_t^2 \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial S_t^2} + \frac{1}{2}\sigma^2 \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial v_t^2} + \frac{1}{2}\sigma_2^2 \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial \theta_t^2} \\
&+ \sigma v_t S_t \rho \frac{\partial^2 U(S_t, v_t, \theta_t, t)}{\partial S_t \partial v_t} + r S_t \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial S} + \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial t} \\
&+ (k(\theta_t - v_t)) \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial v} + \lambda \frac{\partial U(S_t, v_t, \theta_t, t)}{\partial \theta_t} - rU(S_t, v_t, \theta_t, t) = 0.
\end{aligned} \tag{7}$$

When S_t satisfies (2), $U(S_t, v_t, \theta_t, t)$ satisfies

$$\begin{aligned}
U(S_t, v_t, \theta_t, t) &= e^{-r(T-t)} E[\max\{S_T - K, 0\} | S_t], \\
&= e^{-r(T-t)} \int_{\ln K}^{+\infty} (e^g - K) p(g) dg, \\
&= \int_{\ln K}^{+\infty} e^{-r(T-t)} e^g p(g) dg - e^{-r(T-t)} K \int_{\ln K}^{+\infty} p(g) dg,
\end{aligned} \tag{8}$$

where $g = \ln S_T$. Thus, the calculation of $U(S_t, v_t, \theta_t, t)$ can be transferred to finding the probability density function of g, $p(g)$.

Define $f_j (j = 1, 2)$ to be two characteristic functions and $RE[\cdot]$ to take the real part inside the brackets, with i denoting the imaginary unit. It can be shown [21] that the second term on the RHS of (8) can become

$$e^{-r(T-t)} K \int_{\ln K}^{+\infty} p(g) dg = e^{-r(T-t)} K (\frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} RE[\frac{e^{-i\phi \ln K}}{i\phi} f_2] d\phi) \triangleq e^{-r(T-t)} K P_2. \tag{9}$$

Under the risk-neutral condition, the following relationship should be satisfied

$$e^{-r(T-t)} \int_{-\infty}^{+\infty} e^g p(g) dg = e^{-r(T-t)} E[e^g] = S_t, \tag{10}$$

indicating that $\frac{e^{-r(T-t)} e^g p(g)}{S_t}$ is the density function of the integral of the first term on the RHS of (8). Thus, the first term on the RHS of (8) becomes

$$e^{-r(T-t)} \int_{\ln K}^{+\infty} e^g p(g) dg = S_t (\frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} RE[\frac{e^{-i\phi \ln K}}{i\phi} f_1] d\phi) \triangleq S_t P_1, \tag{11}$$

where $f_1 = \frac{f_2}{S_t}$. Therefore,

$$U(S_t, v_t, \theta_t, t) = S_t P_1 - K e^{-r(T-t)} P_2. \tag{12}$$

Bringing (12) into (7) with some transformations ($x = \ln S_t$, $\tau = T - t$), the following partial differential equation is obtained after rectification

$$\begin{aligned}
&\frac{1}{2} v_t^2 \frac{\partial^2 P_j}{\partial x^2} + \frac{1}{2}\sigma^2 \frac{\partial^2 P_j}{\partial v_t^2} + \frac{1}{2}\sigma_2^2 \frac{\partial^2 P_j}{\partial \theta_t^2} + (k(\theta_t - v_t) + b_j) \frac{\partial P_j}{\partial v_t} \\
&+ \sigma v_t \rho \frac{\partial^2 P_j}{\partial x \partial v_t} + (r + u_j v_t^2) \frac{\partial P_j}{\partial x} + \lambda \frac{\partial P_j}{\partial \theta_t} + \frac{\partial P_j}{\partial t} - r P_j = 0.
\end{aligned} \tag{13}$$

Referring to Heston [12,26], it is clear that P_j and f_j must satisfy the same partial differential equation, and the initial value condition for f_j is

$$f_j(x, v, \theta_t, 0; \phi) = e^{i\phi x}.$$

To obtain the characteristic function, f_1 and f_2 are assumed to follow the following functional form [27,28]:

$$f_j = e^{C(\tau;\phi) + D(\tau;\phi) v_t^2 + E(\tau;\phi) v \theta_t + F(\tau;\phi) \theta_t^2 + G(\tau;\phi) \theta_t + H v + i\phi x}, \quad j = 1, 2, \tag{14}$$

with

$$\begin{aligned}
&D(0;\phi) = 0, E(0;\phi) = 0, \\
&F(0;\phi) = 0, G(0;\phi) = 0, \\
&H(0;\phi) = 0, C(0;\phi) = 0.
\end{aligned} \tag{15}$$

Substituting (14) into (13), the ordinary differential equations can be derived as presented in Eq. (4). □

Table 1
Assigned parameter values.

Variable	σ	σ_2	ρ	r	k	$\bar{v}$	λ	θ_0	v_0
Stein–Stein model	0.2	\	0	0.01	5	0.18	\	\	0.15
New model	0.2	0.01	−0.2	0.01	5	\	0.9	0.18	0.15

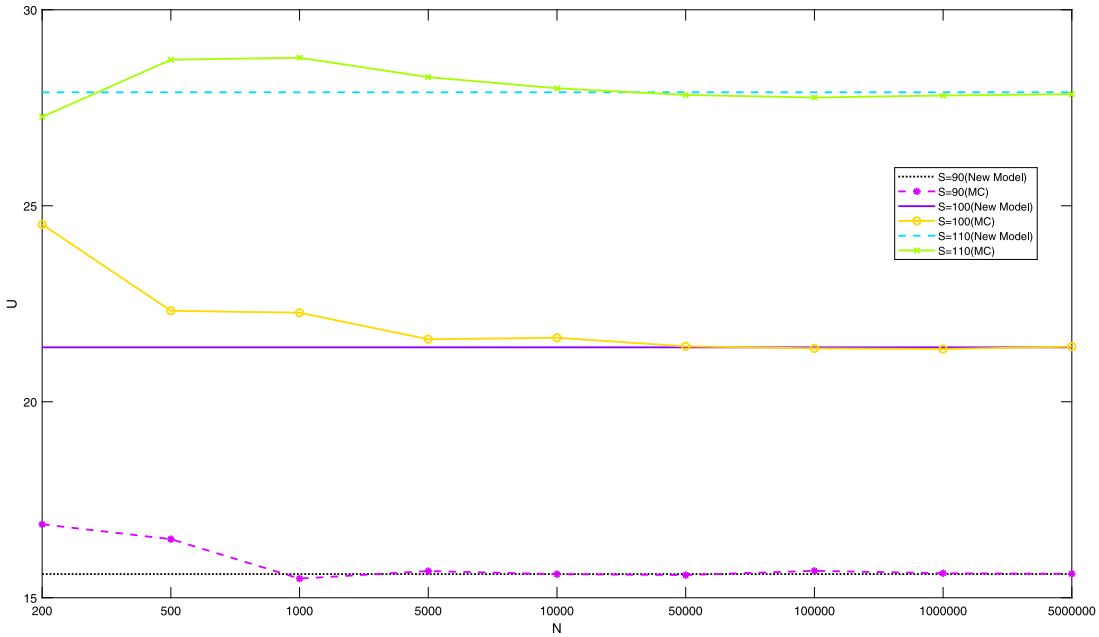


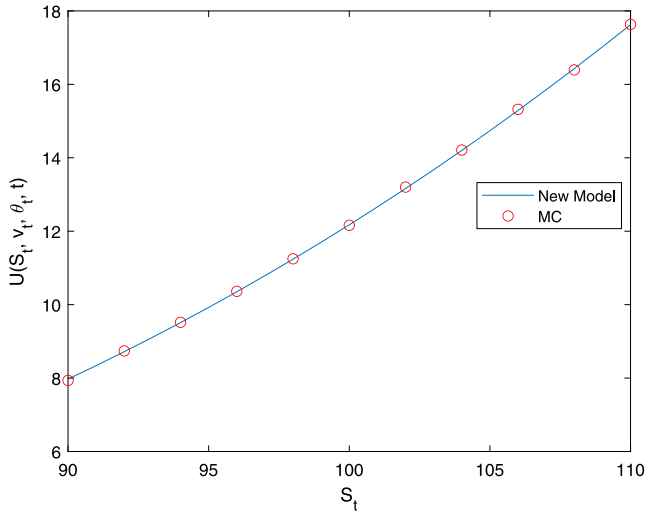
Fig. 1. Relative errors change with the simulation number change.

When calculating the option price, ϕ can be regarded as the outer integral. As ϕ tends to infinity or 0, P_j also tends to 0, which enables P_j to be solved in a finite region using the trapezoidal formula. Using Matlab's ode45 function, the characteristic equation f_j can be derived for any given ϕ. This leads to the solution of P_j, and ultimately allows for the calculation of the price of the call option, represented by $U(S_t, v_t, \theta_t, t)$.

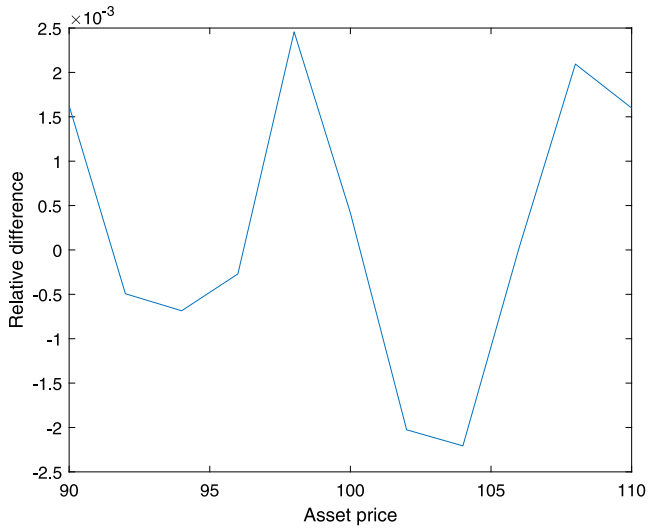
3. Numerical simulations

To compare the performance of the new model with the Stein–Stein model, numerical experiments are carried out by selecting reasonable parameter values according to their own domain, with the magnitude being consistent with the literature [29,30]. Moreover, to facilitate such a comparison, we assume that $\bar{v} = \theta_0$, which means that the long-term average variance of the Stein–Stein model equals the initial long-term average variance of the new model. The current moment (t) is 0, and the specific parameter values are assigned in Table 1. Additionally, unless otherwise specified, this section will evaluate the impact of parameter changes on the pricing of European call options when $\tau = 1$.

From Fig. 1, we can conclude that with the increases in the number of sample paths, denoted by N, European option prices from the two-factor Stein–Stein model get closer to the Monte Carlo (MC) prices. We can find that the prices from the two approaches become very close to each other when the number of simulations is 1,000,000, and thus MC prices of the rest section will be computed with 1,000,000 sample paths. In Fig. 2(a), our model's option price overlaps with that of the MC simulation, demonstrating a positive correlation with underlying prices. Economically, the exercise of the option will only be performed when the underlying asset price exceeds the predetermined delivery price. As the option buyer can earn a premium, the option price is consequently higher. Fig. 2(b) provides clear visual evidence that the maximum relative difference between the pricing results of the two models is only 0.2457%. This indicates that our proposed method for pricing European call options is very close to the MC result. It is worth noting that for the same parameter assignment, the MC method requires 127.12854 s to generate the pricing result, while the new model takes only 5.18182 s. This means that our model is computationally more efficient than the Monte Carlo simulation method.



(a) Option prices.



(b) Relative errors.

Fig. 2. A comparison of the results from the analytical solution and MC simulations.

In Fig. 3, it is evident that the Stein–Stein model produces option prices lower than the new model does. This observation can be attributed to the new model assuming the long-term average variance to be an increasing function of time ($\lambda > 0$) in the numerical analysis. Consequently, the long-term average variance in the new model is higher than that in the Stein–Stein model, resulting in increased volatility and an augmented likelihood of future price hikes and profitability of the option. These factors eventually influence the option prices, leading to the observed disparity. It is noteworthy that this relationship between the models is not universal, and the choice of parameters plays a crucial role. Notably, when λ, θ_2 and ρ all equal zero, the new model converges to the Stein–Stein model. Therefore, this paper primarily focuses on a comparison of different λ values for option pricing across the two models.

In Fig. 4, λ only has an impact on the pricing of options for the new model as the Stein–Stein model has λ as zero. Furthermore, an intersection of the two models demonstrates that a specific parameter combination results in identical pricing outcomes between the new and Stein–Stein models.

The visualization in Fig. 5 reveals a positive correlation between the pricing of options according to the new model and both σ_2 and λ. Furthermore, it is worth noting that changes in σ_2 have a more significant impact on option pricing than changes in λ. To account for this phenomenon, one can explain from the point that σ_2 captures the uncertainty surrounding the long-term average variance, while λ denotes the expected value of that mean. As such, a climb in σ_2 yields an even greater increase in the long-term

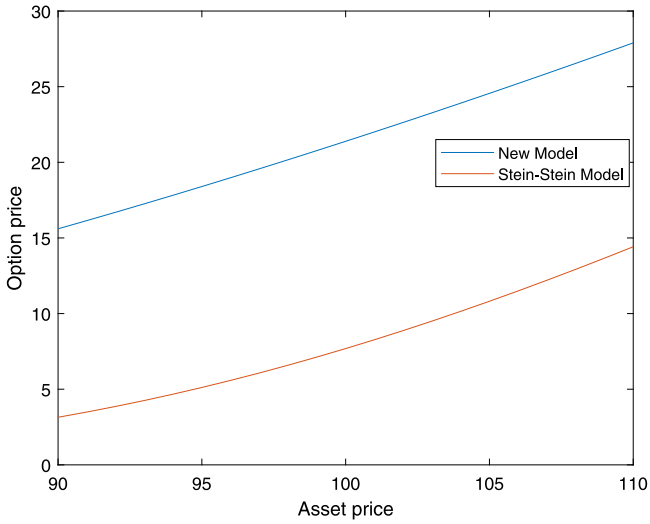


Fig. 3. A comparison of two models with different S_0.

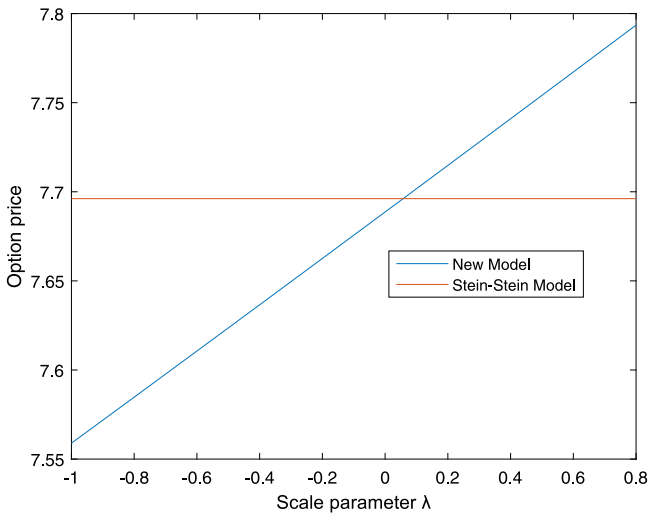


Fig. 4. A comparison of two models with different different λ.

average variance, which, combined with the positive correlation between asset volatility and its long-term average, leads to greater price volatility for the asset. This increased volatility results in a greater rise in S_t, and subsequently, the option price increases as well.

To provide a numerical illustration of the correlation between volatility and European call option price (ρ) mentioned previously and its impact on the model's pricing, we take θ_0 as the horizontal coordinate and vary ρ to derive the option price curve, as shown in Fig. 6. Our analysis reveals that, regardless of the magnitude relationship of ρ, the model pricing increases when θ_0 rises. This relationship between θ_0 and option pricing can be traced back to the relationship between volatility and option pricing explained for volatility in Fig. 5. It should be noted that the effect of changes in ρ as θ_0 changes is inconsistent. As θ_0 approaches 0, the option price decreases as ρ increases, whereas as θ_0 approaches 1, the option price rises as ρ increases. The convergence towards 0 for θ_0 indicates that the long-term average variance is falling, while the fact that ρ is greater than 0 implies a positive correlation between asset and volatility. All things being equal, declining volatility leads to a decrease in option prices. Furthermore, the convergence towards 1 or vice versa is consistent with the real-world observation that risk and return are positively correlated.

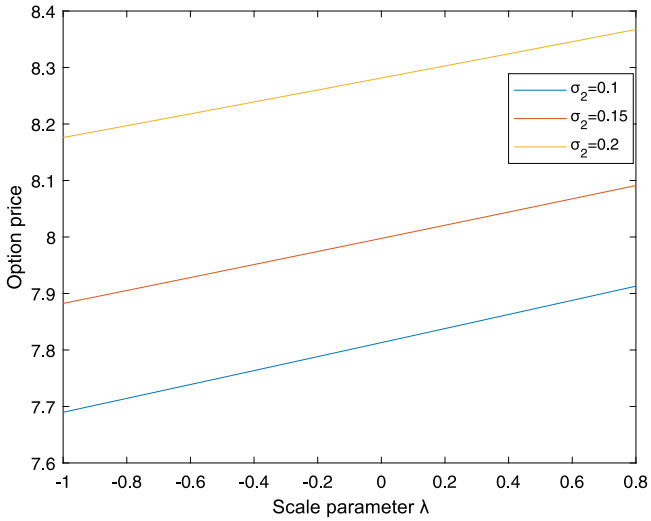


Fig. 5. The effect of σ_2 and λ.

4. An empirical study

4.1. Data selection

Given the exponential growth of the Chinese securities market in recent years and the escalating involvement in financial derivatives trading, there has been a notable surge in the volume of accessible data. Therefore, to test the applicability of the new model in the emerging derivative markets like the Chinese derivative market, we choose the SSE 50 ETF option and its underlying fund as the data sources.

It is worth noting that, for this study, the daily high and low prices of the unadjusted SSE 50 ETF call option and its underlying asset are averaged to obtain the prices for the call option and the underlying asset, respectively. Data screening is carried out using R language, and for the sake of data continuity, the study selects data related to the call option price every Tuesday over eight months spanning from October 12, 2021 to June 7th, 2022. This eight-month period enables the estimation of the model's seven parameters, namely θ_1, θ_2, ρ, v_0, k, θ_0 and λ resulting in 32 sets of parameter estimates.

The parameters estimated on Tuesday are applied as model parameters for predicting the call option price alongside data from the following day (Wednesday) or the third day (Thursday). The predicted results are then compared against the actual call option price on the forecast day to examine the error in the two models, referred to as out-of-sample error.

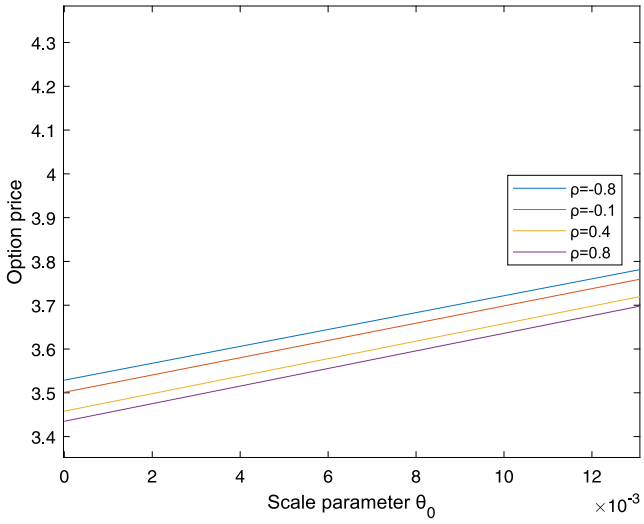
4.2. Parameter estimation

This paper's estimation of the model's seven parameters is a nonlinear estimation problem. Traditional parameter estimation methods may result in local optimal solutions. The genetic algorithm is a more widely used method of parameter estimation as it is more robust and precise. For the sake of accuracy, this paper uses Matlab's built-in genetic algorithm to derive the necessary parameters for the model.

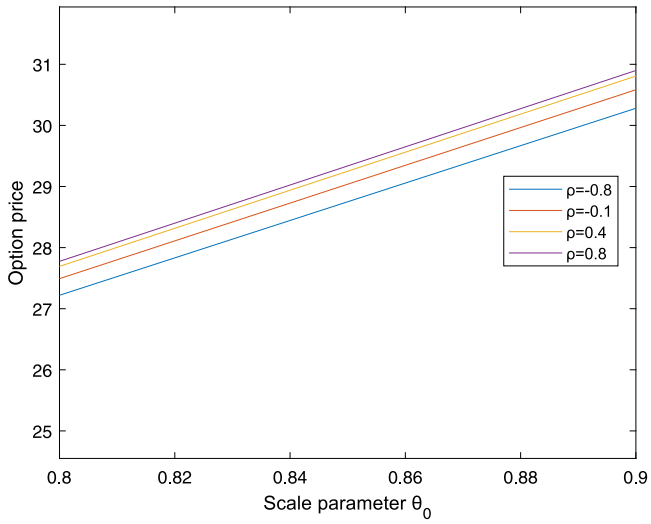
When it comes to parameter estimation, the two primary objective functions to choose from are Percent Mean Squared Error (PMSE) and Mean Squared Errors (MSE) [31]. As the PMSE method can accommodate a greater proportion of options with low prices, this paper opts to use the MSE method to estimate the necessary parameter values. The estimated parameters are then used to predict the expectations for call option prices.

$$\mathrm{MSE} = \frac{1}{N}\sum_{k=-1}^{N}(U_i^{Market} - U_i^{Model})^2 \tag{16}$$

Where U_i^{Market} denotes the real call price and U_i^{Model} denotes the estimated call price. In this paper, we take the parameters with the smallest MSE as the optimal solution.



(a) Option price for different values of ρ when θ_0 tends to 0



(b) Option price for different values of ρ when θ_0 tends to 1

Fig. 6. Option price for different values of ρ and θ_0.

4.3. Model performance

The genetic algorithm was utilized to obtain two sets of parameter estimates for each Tuesday in the sample (see Table 2[1]). As evident from these results, $\rho > 0$, indicates an inverse leverage effect of volatility; that is to say, when prices are high, volatility increases and future changes in the price of the underlying asset go up. This finding is consistent with a study by Qiu et al. [32] focusing on the Chinese index market.

The data presented in Table 2 is combined with the values of S, K and τ for Tuesday to derive U_i^{Model} using the analytical solution of call option pricing. The calculated MSE for each day is depicted in Fig. 7. The figure indicates that the errors produced by the new model are lower than those generated by the Stein–Stein model, with mean values of 1.968×10^{-4} and 2.340×10^{-4} respectively, which indicates that there is an improvement of 16% in the accuracy when the new model is used.

[1] Note: The data in the table are derived by taking the mean values of the parameter estimates after parameter estimation using data from the sample period for each of the two models.

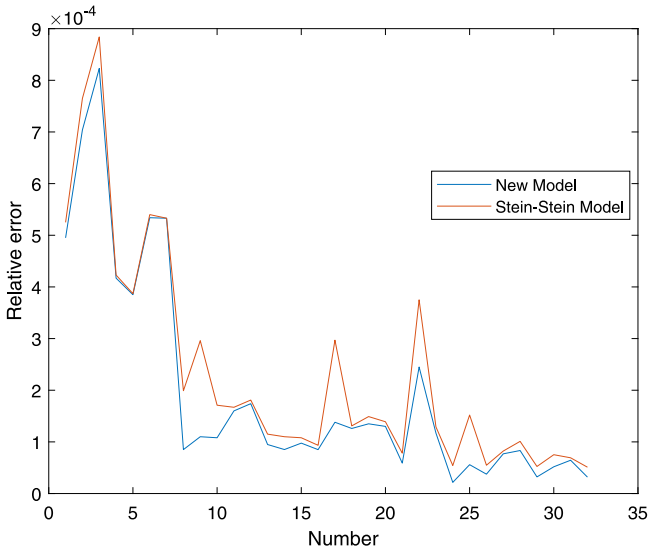


Fig. 7. Error comparison of the two models.

Table 2
Estimated parameters.

Variable	σ	σ_2	ρ	k	$\overline{v}$	λ	θ_0	v_0
Stein–Stein model	0.5593	\	0	3.8111	0.1459	\	\	0. 2159
New model	0.3943	0.4112	0.1691	4.9394	\	0.3115	0.1319	0.2924

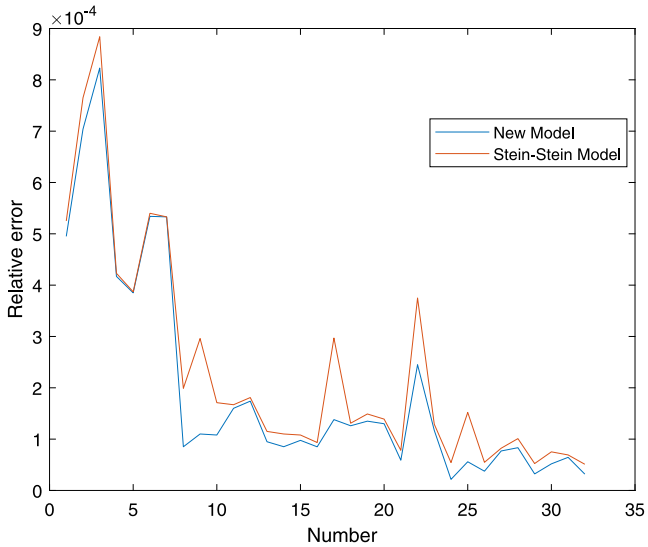
Table 3
The significance tests of out-of-sample errors under both models.

	Wednesday	Thursday
t-test (P value)	1.405×10^{-04}	3.300×10^{-03}

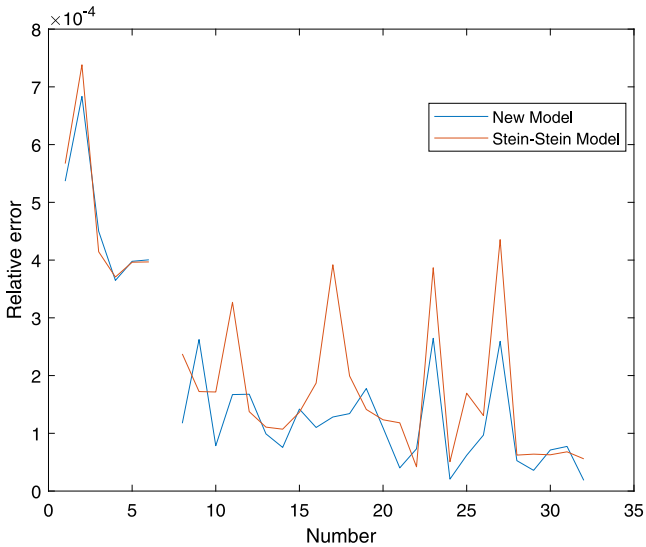
The estimated parameter values are used to predict the call option price for Wednesday (Thursday) by combining the values of S, K and τ for that day. The resulting daily errors are illustrated in Fig. 8. In this figure, it is evident that the pricing bias is comparatively high for the first seven groups of options. This could be because the option prices during the November holiday period were anomalous and unable to reflect the latest market information. Post the holiday period, the new model performed more consistently and had pricing errors within 5×10^{-4}. Analysis of the new model's prediction of the next day's option prices (Fig. 8(a)) shows it outperforms the Stein–Stein model, albeit with slightly higher pricing errors in certain groups. Nevertheless, Fig. 8(b) reveals[2] that the new model has less pricing error than the Stein–Stein model for next-day option prices in most cases. Additionally, the mean value of Wednesday (Thursday) option price bias produced by the new model is approximately 16% (20%) lower than that of the Stein–Stein model. Therefore, it can tentatively be concluded that the out-of-sample errors from the new model stay below those under the Stein–Stein model overall.

To further ascertain whether the pricing errors of the two models were significantly different, a t-test was conducted, and the results are presented in Table 3. As per the table, it is apparent that the pricing errors of the Tuesday parameter estimates for Wednesday (Thursday) options prices differ significantly between the two models at a 1% significance level. This confirms that the out-of-sample performance of the new model is generally superior to that of the Stein–Stein model, thereby validating the necessity and reasonableness of the new model as an improvement over the Stein–Stein model. One of the main reasons for this phenomenon is that with the introduction of λ and σ_2, the new model can capture the implied volatility of the option better, which can be drawn from Fig. 11 to be presented later.

[2] The seventh group of data in this figure is not available because the option data on November 25 are all adjusted data and are not part of the study selected for this paper.



(a) Relative error when estimating Wednesday option price



(b) Relative error when estimate Thursday option price

Fig. 8. Mean value of out-of-sample errors of the two model.

To specifically analyze the components of the forecast Wednesday option price error, first define the degree of moneyness of a call option as $\frac{S_t-K}{K}$, where S_t, K are defined in the same way as before. Call options with values greater than 10% are classified as in-the-money options, values less than −10% are classified as out-of-money options, and calls with other values are classified as at-the-money options. The pricing errors for each group of options with different degrees of moneyness are shown in Fig. 9.[3] From this figure, it can be seen that the new model has the most accurate prediction for out-of-money options and a slightly worse fit for in-the-money options in the first 7 groups of predicted data. It can be seen that the errors in the first 7 sets of data of Fig. 8(a) are mainly composed of the errors of the in-the-money options.

[3] The discontinuity in the real option pricing error curve in the figure is because the real option condition as defined in this paper is not satisfied in the unadjusted European call option data traded on the same day.

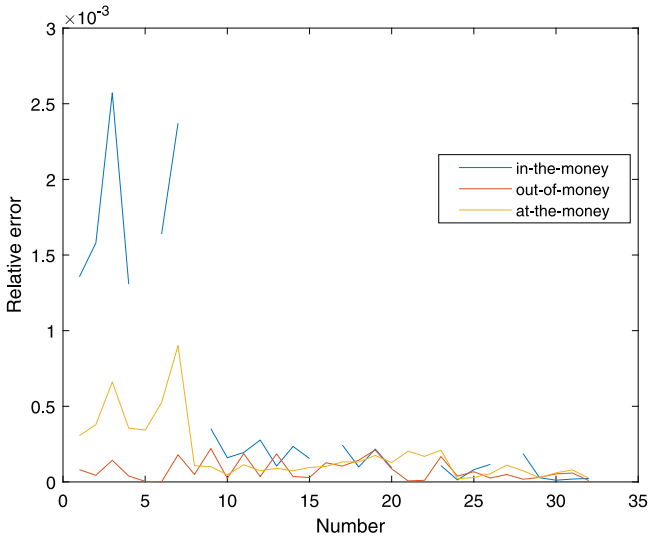


Fig. 9. Errors across moneyness.

Table 4
Out-of-sample errors of both models across moneyness.

	out-of-money	at-the-money	in-the-money
Stein–Stein model	1.220×10^{-04}	2.673×10^{-04}	3.519×10^{-04}
New model	1.239×10^{-04}	2.408×10^{-04}	2.277×10^{-04}

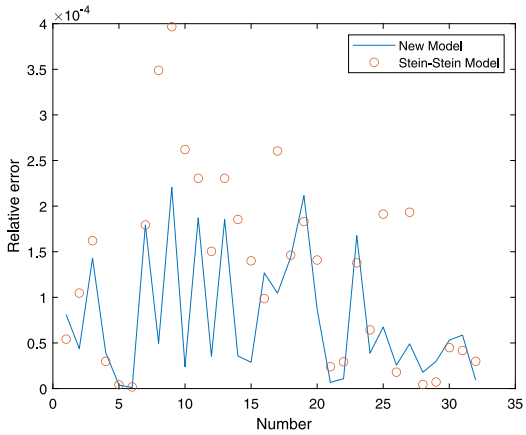
By comparing the new model with the Stein–Stein model in terms of pricing options with different moneyness in Fig. 10,[4] it is evident that the new model is more accurate in pricing out-of-money options for most groups. Additionally, the new model outperforms the Stein–Stein model significantly in pricing at-the-money options. However, the new model has more groups for which the in-the-money option pricing bias is larger than that of the Stein–Stein model.

Table 4 presents an additional mean error comparison of both models across different moneyness. The results demonstrate that, overall, the new model outperforms the Stein–Stein model for both at-the-money and in-the-money options.

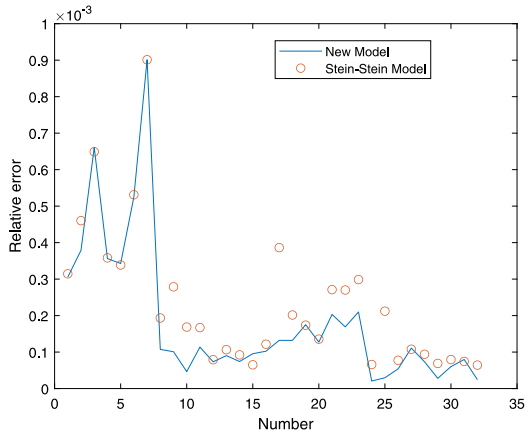
Furthermore, when analyzing option pricing models, researchers often examine the fitting effect of implied volatility. To determine the implied volatility of both models and the market, we utilized the blsimpv function in Matlab. Our findings demonstrate that the implied volatility of the new model is more closely aligned with the market-implied volatility than the implied volatility of the Stein–Stein model. Fig. 11(a) provides an example of the implied volatility within the actual option prices and estimated option prices for the 24th data set, with 80 days to expiration, across different delivery prices. On that day, the market price of the SSE 50 ETF fund was 2.8380. The lowest implied volatility for both pricing models was around that value, suggesting that both models are effective in explaining the "smile curve of volatility" phenomenon. However, differences arise in the changing trends of volatility and the error of volatility values. The implied volatility curve of the new pricing model matches with the market-implied volatility curve, whereas the Stein–Stein model's implied volatility is smaller than that of the new model only when the delivery price of SSE 50 ETF options is at around 3.3. When the delivery price is lower than 3.3, the Stein–Stein model's volatility considerably deviates from the market-implied volatility.

To mitigate the influence of chance, a group of options with various delivery prices and 234 days left in the expiration period was selected to compare the model-implied volatility with the market-implied volatility, as shown in Fig. 11(b). Clearly, the shape of market-implied volatility can almost be replicated by implied volatility produced by the new model, while implied volatility generated by the classical Stein–Stein model provides very poor fitting results. This can actually explain the phenomenon that the new model considering a stochastic long-term mean can provide better match to market data, compared with the classical Stein–Stein.

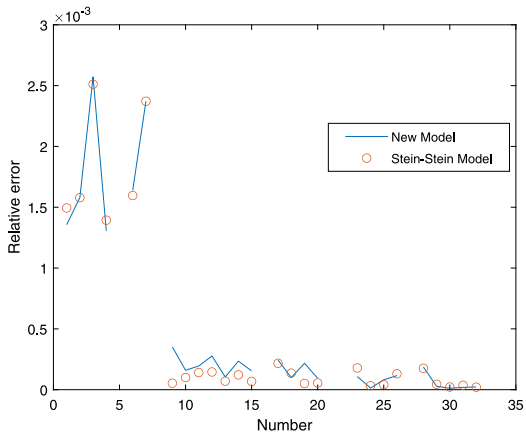
[4] Note: The discontinuity in the curve in Fig. 10(c) is because the in-the-money option condition as defined in this paper is not satisfied in the unadjusted European call option data traded on the same day.



(a) Pricing errors of out-of-money options in two models



(b) Pricing errors of at-the-money options in two models



(c) Pricing errors of in-the-money options in two models

Fig. 10. Pricing errors of difference degree of moneyness options in two models.

5. Conclusion

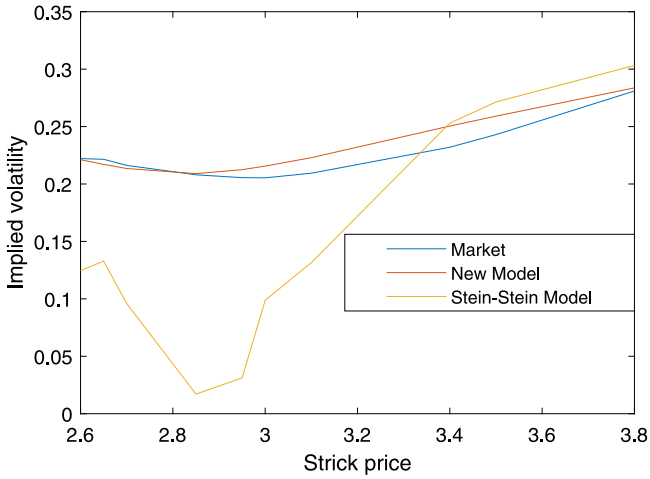
This paper introduces a new model by relaxing two significant assumptions (the volatility of the model is uncorrelated with the underlying asset, and that the long-term mean remains constant) made in the Stein–Stein model. The new model yields pricing results almost identical to those of the Monte Carlo simulation, while outperforming the simulation in terms of solution rate. Furthermore, it is demonstrated that the new model accurately captures realistic SSE 50 ETF prices and its market-implied volatility, showcasing superior performance over the Stein–Stein model.

Data availability

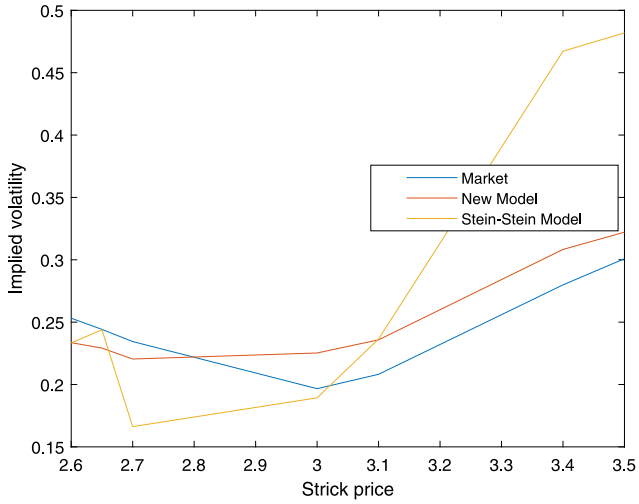
Data will be made available on request.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (No. 12301614, No. 12101554) and the Fundamental Research Funds for Zhejiang Provincial Universities (No. GB202103001). The authors would like to gratefully acknowledge the anonymous referees' constructive comments and suggestions, which greatly help to improve the correctness and readability of the manuscript.



(a) 80 days to expiry.



(b) 234 days to expiry.

Fig. 11. Implied volatility.

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